

# Texture-based Feature Extraction and Machine Learning Model for the Detection of Acute Lymphoblastic Leukemia

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**Abstract**— Acute Lymphocytic Leukaemia (ALL) is a dangerous malignancy that not only attacks adults but also contributes to roughly 25% of children. A precise and on-time diagnosis of cancer is a critical prerequisite for effective therapy to enhance survival rates. It is challenging to identify cancer cells from normal cells because leukemic cells seem extremely identical to normal cells beneath the microscope. Normally, morphological analysis for the detection of ALL is conducted physically on blood cells by qualified medical individuals, which has various drawbacks, like a lack of medical experts, slow analysis, and prognosis which depends on the knowledge of medical professionals. For this reason, we provide a robust method for improving the reliability of ALL identification using feature extraction and a Machine Learning (ML) model to discriminate between cancerous and healthy cells in images. To extract texture features for analysis, we employ the First Order Histogram (FOH) and the Gray-Level Co-occurrence Matrix (GLCM) in our technique. Logistic Regression (LR), Naive Bayes (NB), and Artificial Neural Network (ANN) are used in the categorizing method. The simulation results demonstrate that combining GLCM feature extraction with the ANN model yields the highest accuracy, correctly predicting 595 out of 600 data samples. The accuracy was 99.17%, which is quite remarkable.

**Keywords**— *Leukemia, Microscopic Images, Texture Features, Artificial Neural Network, Confusion Matrix*

## I. INTRODUCTION

ALL, Chronic Lymphoid, Chronic Myeloid, and Acute Myeloid Leukemia (AML) are the various kinds of leukemia that affect the bone marrow [1]. ALL is the most

prevalent kind of cancer in children and the second most frequent type of adult acute leukemia. Despite a 90% success rate in treating ALL in children, it remains a top cause of mortality in both children and adults. Early diagnosis allows for faster treatment and improved survival rates. Depending on the patient's symptoms and risk level, chemotherapy, radiation, anticancer drugs, or a combination of these may be used to treat cancer [2]. Only after a comprehensive diagnosis can the appropriate therapy be selected, which necessitates the use of time-consuming and arduous manual methods done by skilled medical staff. The microscopic examination of blood cells by expert pathologists is critical in discovering abnormal patterns [3].

Despite its importance in diagnosis, the interpretation of medical images is plagued with error due to variables such as tissue variation, a scarcity of experts, lengthy working hours, eye weariness, and focus issues. In order to mitigate the severity of these issues, research is being conducted to automate the interpretation of medical images. Deep learning and ML advancements have enabled computer-assisted image analysis to play a role in early disease detection. Doctors and pathologists can use advanced algorithms to examine blood slides for symptoms of ALL. The purpose of this study is to use texture-based feature extraction and ML algorithms to analyse microscopic images of patients in order to swiftly and reliably identify and categorise them as having ALL. By simplifying operations, this unique technique promises to enhance diagnostic accuracy and facilitate treatment.

## II. RELATED WORKS

We've summarised some of the papers we studied throughout our research in this section. In the study [4] the author introduces a unique and computationally efficient technique for the automated detection and classification of ALL and AML. The Transfer Learning (TL) approach is chosen as a foundation because of its shown success in medical image analysis with few datasets. The suggested approach employs a lightweight TL-based feature extraction approach in conjunction with a Support Vector Machine (SVM) to increase the detection system's speed and efficiency. A depth-wise separable convolution, an adjustable multiplier, and an inverted residual bottleneck structure are notable aspects that contribute to the system's computational efficiency. The SVM-based classification improves optimization even more by pinpointing the precise position of the hyperplane. According to experimental results, the proposed technique outperforms ALL and AML detection on three publicly available gold-standard datasets. A study [5] looks at how to identify Acute Myeloid Malignancy (AMM) using microscopic images of white blood cells. Preparation is stressed, with an emphasis on using morphological methodologies to separate white blood cells from blood smear images. The separated sections are then examined for a comprehensive set of geometrical, textural, and statistical properties. This study uses four distinct ML techniques to link acute lymphoblastic cells to leukemia malignancy. The SVM is the most successful. Despite the challenges posed by the wide range of blood smear images, the EMC-SVM successfully differentiates and classifies white blood cells in sample blood smear images. Using the provided method, each segmented cell is consistently placed in the relevant category. The study [6] uses EfficientNet, a cutting-edge and robust deep learning model, to create a simple and effective TL-based ALL detection method. The efficiency of eight distinct variants of EfficientNets utilized for feature extraction is evaluated using classification accuracy. The study's excellent results stem from the employment of an ensemble of three cutting-edge classifiers: the SVM, the Random Forest (RF), and the LR. The efficacy and accuracy of the proposed system in ALL detection may be due to the use of EfficientNet in combination with the ensemble approach.

Article [7] describes efforts to enhance diagnostic speed and accuracy by automating the identification of leukemia in peripheral blood using artificial vision techniques. Unlike previous efforts, this study attempts diversity by applying the recommended technique to datasets from several sources. The linear scaling filter to reduce noise, global-local contrast enhancement to improve image quality, watershed, and K-means clustering are employed to segment it. The cell, nucleus, and cytoplasm sub-images are further subdivided into 103 distinct properties. The characteristics are then reduced to the top 50. A highly accurate, trained SVM model is used in the investigation.

The approach is encapsulated in an executable MATLAB script, and its ability to detect leukemia in a new dataset demonstrates that the method is functional despite changes in the underlying images. The study's [8] goal is to provide diagnostic tools for the early detection of leukemia using two distinct ALL Image Databases. The images are pre-processed with average and Laplacian filters before being fed into the three recommended systems. The first system is artificially intelligent and uses a mix of ANN, Feed-forward Neural Network (FFNN), and SVM for feature extraction. ANN and FFNN are nearly equivalent in terms of accuracy. The second system employs TL-based convolutional neural network (CNN) models AlexNet, GoogleNet, and ResNet-18. Deep feature map extraction and categorization enable these models to achieve high accuracy. The third system combines CNN models with SVM classification to generate feature maps. All three hybrid techniques show promising results in detecting leukemia at an early stage in these data sets. In the article [9], they present a supervised ML model that can predict the likelihood of early-stage leukemia based solely on symptoms, and they explain how it works. The primary data originates from two big hospitals in Bangladesh, and it comprises sixteen characteristics obtained via an expert consultation poll. This open-source supervised model beats rival algorithms by employing a decision tree classifier and creating rules that everyone can understand. The Apriori approach is used to construct these rules, which enhances the model's interpretability for forecasting instances of leukemia. Furthermore, feature analysis and selection are performed to highlight the importance of specific traits and improve the overall efficacy of classification models. The proposed model is the simplest of those examined, but it consistently beats the other classifiers. The decision tree model, in particular, performs well in the test set evaluation, confirming the method's potential for developing accurate and interpretable predictions of early-stage leukemia from symptom data alone.

## III. MATERIALS AND METHODS

The ability to detect leukemia in its earliest stages is essential for effective treatment. Automatic identification requires the following procedures:

- Leukemia and healthy tissue samples are gathered from the standard dataset.
- Pre-processing operations, such as resizing and normalization, are performed on the obtained data.
- Two independent strategies are applied for feature extraction: GLCM and FOH. Seven features are produced by GLCM that capture spatial connections in the image, whereas six features are produced by FOH that characterize the intensity distribution.
- The collected features from GLCM and FOH are independently fed into three ML models.
- The combination of the feature extraction technique and ML model that yields the best performance is identified for automatic leukemia identification. Figure 1 illustrates the procedure using the flow chart.

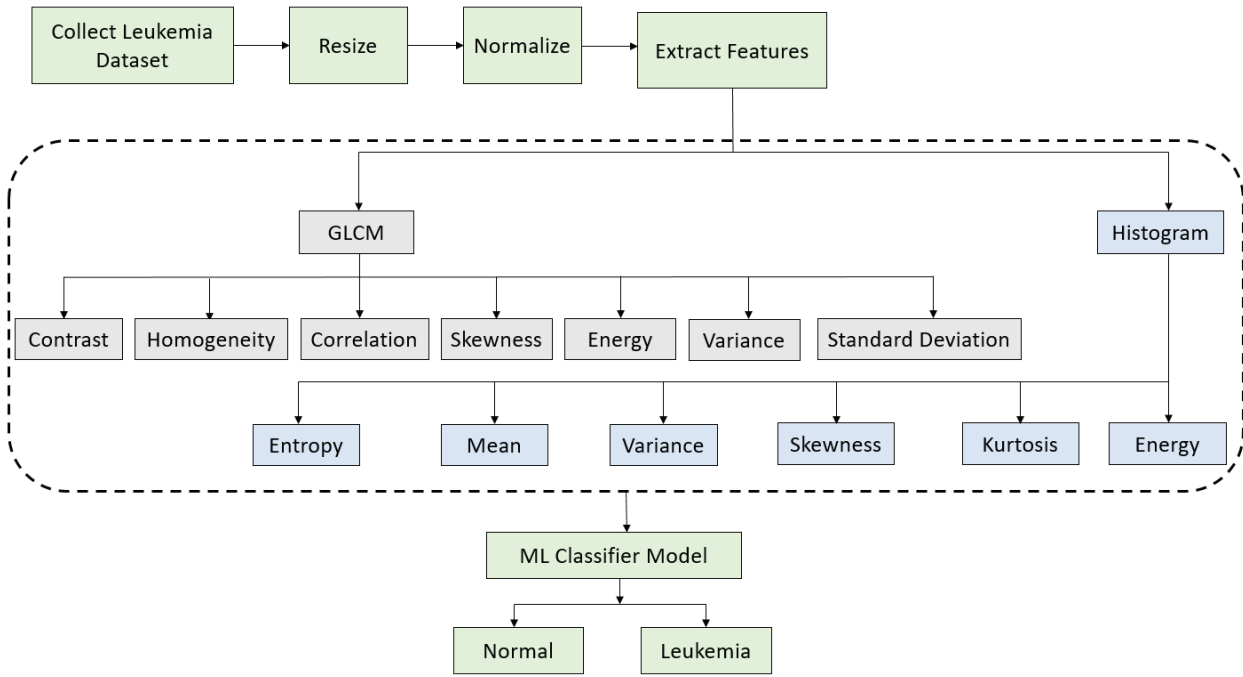


Fig 1. Workflow of Leukemia detection

#### A. Acquisition and Pre-processing of Dataset

According to [10], the C-NMC 2019 dataset is a large compilation of 15,114 lymphocyte images from 118 individuals. There are three separate directories in this collection, each with its own function. There are 10,661 cells total, with 3,389 healthy cells from 26 people and 7,272 malignant cells from 47 subjects housed in the "training" folder. There are a total of 1,867 cells in the "validation" folder, including 1,219 cancerous cells from 13 patients and 648 healthy cells from 15 patients. A total of 2,586 unlabelled cells from 17 individuals may be found in the

"testing" folder. Notably, these files contain individual cell images of both cancerous and benign cells, skilfully diagnosed by seasoned oncologists. The distribution of both raw and taken leukemia data is further shown in Table 1. Essential pre-processing processes are performed on the gathered data to assist further analysis and model training. Initially, the images are downsized to a standardized dimension of 256x256 pixels, and then, the pixel values are normalized to a range between 0 and 1. These pre-processing steps are crucial for making the dataset research-ready for in-depth analysis and the creation of efficient analytical models.

Table 1. Leukemia data distribution

|                      | Train    |         | Validate |         | Test            |         |
|----------------------|----------|---------|----------|---------|-----------------|---------|
|                      | Leukemia | Healthy | Leukemia | Healthy | Unlabelled Data |         |
| <b>Raw Data</b>      | 7272     | 3389    | 1219     | 648     | 2586            |         |
|                      | Leukemia | Healthy | Leukemia | Healthy | Leukemia        | Healthy |
| <b>Balanced Data</b> | 2000     | 2000    | -        | -       | 300             | 300     |

#### B. Extract Features

**FOH:** The FOH feature extracts fundamental statistical textural information from an image [11]. The image's two spatial variables,  $x$  and  $y$ , are represented by numbers ( $i = 1, 2, 3, \dots, X-1$  and  $j = 1, 2, 3, \dots, Y-1$ ). This notation effectively captures the intensity distribution of levels for the function  $(i, j)$  with discrete values, where  $i$  can take values ranging from 0 to  $N-1$ , and  $N$  represents the total intensity levels. The histogram is a precise graphical representation that provides 1<sup>st</sup> order statistical data about the image by displaying the intensity value distribution. The shape of the histogram, which may be symmetric, bimodal, or multimodal, offers information on the nature of the image.

The textural qualities obtained from first-order histograms are determined by the formulae below.

$$Mean = \sum_{i,j=0}^{N-1} i * P_{ij} \quad [1]$$

$$Variance = \sum_{i,j=0}^{N-1} (1 - \mu)^2 * (P_{ij})^2 \quad [2]$$

$$Skewness = \sum_{i,j=0}^{N-1} (1 - \mu)^3 * P_{ij} \quad [3]$$

$$Kurtosis = \sum_{i,j=0}^{N-1} (1 - \mu)^4 * (P_{ij}) - 3 \quad [4]$$

$$Energy = \sum_{i,j=0}^{N-1} (P_{ij})^2 \quad [5]$$

$$Entropy = - \sum_{i,j=0}^{N-1} P_{ij} * \log P_{ij} \quad [6]$$

And calculate the probability density of the intensity level, indicated by  $P_{ij}$ , by dividing the intensity at a pixel by the number of pixels in the image. The variance represents the intensity fluctuation around the mean, indicating the image's roughness, whereas the mean represents the average intensity level throughout the image. Skewness evaluates the histogram's symmetry, whereas kurtosis gives information on the accumulation of data around the mean. Entropy is the calculation of information content that has an inversely qualitative relationship with energy. It relates to the intensity values' uniform distribution. When energy levels are low, entropy levels are high. The nucleus peak in a normal cell's histogram is substantially smaller than the cytoplasm and background peaks due to its tiny size. An aberrant cell's histogram will contain three peaks, one for each, instead of two for the nucleus and cytoplasm, as normal cells do. When healthy cells transform into cancerous ones, the nucleus expands. Normal cells have a higher positive skewness value than aberrant cells, which have short tails in the left direction.

**GLCM:** Haralick et al. [12] proposed an extremely successful texture description. Each segment  $(i, j)$  of the GLCM matrix indicates a specific pair of pixel occurrences. The intensity data of the segmented image was used to determine four texture characteristics: contrast, correlation, energy, and homogeneity. One of these qualities, which reflects the range of local intensity changes in an image, is contrast, which is calculated as the difference between the maximum and least intensity values.

$$CONT = \sum_{i,j} |i - j|^2 A_{ij} \quad [7]$$

Homogeneity measures the smoothness of reduced-level variations in segments. Higher homogeneity scores imply a smoother distribution of intensity in the image and so correspond to lower contrast levels.

$$H = \sum_{i,j} \frac{A_{ij}}{1 + |i - j|} \quad [8]$$

The degree to which pixels in an image are connected to one another is measured by the correlation feature.

$$CORR = \sum_{i,j} \frac{(i - \mu_i)(j - \mu_j)A_{ij}}{\sigma_i \sigma_j} \quad [9]$$

Skewness measures the degree of divergence of intensity values from the standard distribution. The pixel allocation range is evaluated between 0 and -1 to +1, providing information on the asymmetry and shape of the image.

$$SK = \sum \frac{((GLS - \mu_{GL})^3 * PixelCount)}{(NumberOfPixels - 1)^3 * \sigma^3} \quad [10]$$

Skewness is affected not just by the number of data points to the left and right of the mode, but also by their distance from it. As a result, a greater number of data points to the left of the mode might not be more meaningful than a greater number of data points to the right of the mode. As a result, despite the fact that there are more data points on the left, the total skewness is positive, indicating a bias to the right.

The energy of the entire intensity distribution in the image can be quantified by summing the squared components.

$$E = \sum_{i,j} (A_{ij})^2 \quad [11]$$

The variance, computed as the mean squared deviation, indicates how much intensity levels fluctuate around the mean.

$$Var = \frac{1}{n} \sum_{i=1}^n |X_i - \mu|^2 \quad [12]$$

The standard deviation, which is equal to the square root of the variance, may be used to calculate the degree of dispersion or variation in the intensity values.

$$Std = \sqrt{\frac{1}{n} \sum_{i=1}^n |X_i - \mu|^2} \quad [13]$$

### C. ML Classifiers

**LR:** LR is a well-established technique for classification that is a type of multiple regression created particularly for instances where the response is binary rather than quantitative [13]. In the most basic sense, the feature variables are not fully random. The answer, which represents the class, is a binary random variable with the value 1 (identifying the class of interest) and a specified probability given as  $p$ , and the value 0 with a probability of  $1 - p$ . The "success probability"  $p$  is determined by the values of the feature variables, and is calculated as the logarithm of the odds ratio, commonly known as the "log odds." The linear function of the predictor variable is given below and it is critical in this LR design.

$$\log \left[ \frac{p}{1-p} \right] \quad [14]$$

Setting a cut-off value, which is typically 0.5, is a critical step in utilizing logistic regression for classification. If a case's estimated or fitted success probability is greater than (or equal to) the cut-off, the case is categorized as class 1, otherwise it is classified as class 0. Despite the fact that this approach generates a linear classification border, it does not necessarily correspond with the boundary produced by linear discriminant analysis. Variable significance tests can be used to choose a feature in the categorization. Several iterative (step-by-step) variable selection techniques have made their way into current computer systems. Logistic classification has become a popular data mining strategy since it is conceptually similar to multiple regression and benefits from automated variable selection. Another advantage is that, rather than simply predicting a class, it may additionally estimate a probability of success depending on the values of the feature variables. This enables an arbitrary cut-off for classification to be determined (it does not have to be 0.5) and observations to be ranked according to their success likelihood. Despite this flexibility, it is critical to understand that logistic classification inherently contains a linear classification boundary, which limits the classifier's potential utility.

**NB:** The NB classifier works based on the Bayes theorem and posits that each feature has an equal and independent

influence on the target class [14]. It is assumed that each feature independently and consistently increases the likelihood that a sample belongs to a specific class. Because of its simplicity, computational economy, and competent performance on large datasets with high dimensionality, the Naive Bayes classifier is ideal for real-time applications and is resistant to noise. The classifier calculates class probabilities  $P(y_i)$  and conditional probabilities by dividing the frequency of each feature value for a given class value by the frequency of instances having that class value. It is vital to remember that the Naive Bayes classifier performs best when applied to datasets that have had associated features removed. This is because associated traits may be given undue weight in the model, resulting in a duplicate count. Applying the Naive Bayes classifier to training samples with three features  $(X_1, X_2, X_3)$  and a single label  $(y_i)$ , where  $i$  is an integer between one and two, is presented. The classifier's primary purpose is to assign a single label,  $y_1$  or  $y_2$ , to a sample based on its feature values. The training dataset is evaluated in the first phase of the algorithm to determine a conditional probability for a class  $y_i$  given a set of feature values  $(X_1, X_2, X_3)$ .

$$Py_iX_1X_2X_3 = \frac{PX_1y_iX_2y_iX_3y_iPy_i}{PX_1PX_2PX_3} \quad [15]$$

The numerator in Equation (15) is seen to be constant within the context of a given dataset. As a result, we reduce Equation (15) to a proportionality, denoted as:

$$Py_iX_1X_2X_3 \propto PX_1y_iX_2y_iX_3y_iPy_i \quad [16]$$

The dataset is segmented based on classes in order to calculate statistical parameters like mean and variance for the feature distributions. In the next step,  $P\left(\frac{X_j}{y_i}\right)$  is computed without including the other classes, as their parameters are determined separately. After a sample has been collected with known feature values, the method calculates the likelihood that the sample belongs to a specified class. To do this, pick  $y_i$  such that  $P\left(\frac{X_1}{y_i}\right)P\left(\frac{X_2}{y_i}\right)P\left(\frac{X_3}{y_i}\right)$  is as high as possible. The claim may be expressed mathematically as:

$$y = \operatorname{argmax}_{y_i} PX_1y_iPX_2y_iPX_3y_iPy_i \quad [17]$$

The highest probability hypothesis is chosen, and this is known as the maximal a posteriori decision rule.

**ANN:** ANNs are a method of ML and classification that takes its cues from the structure of real brains. They serve as potent classifiers, backed by mathematical evidence stating that an ANN can learn any mathematical function to arbitrary precision given adequate training time and an extensive dataset. Nodes, which are like neurons, and connections, which are like an axon or dendrites, are the building blocks of artificial neural networks. The weights of these associations define them. There are several layers of neurons, with the input layer representing a particular data vector and the output layer providing the classification result. Additionally, there may be one or more hidden layers, so termed because they do not have any connections to the outside world [15]. The perceptron, a type of fully connected feed-forward network, is the most basic and common type of ANN. Although ANNs with a wide variety of node counts

and types of connection are available, perceptron's have found the most widespread use. Each hidden layer and output layer node in such networks uses a weighted sum of its inputs to get its final input value. The output value of a node  $j$  is the total of the input values for that node, which are calculated by multiplying the output value of each connected node  $i$  by the weight,  $w_{i,j}$ , on that connection.

$$x = \sum_{i=1}^n v_i w_{i,j} \quad [18]$$

The resultant output works as an input for the next layer of nodes in the setting of hidden layer nodes. This value represents the ultimate result for nodes specified as outputs. While sigmoid functions are the most commonly utilized as transfer functions, any nonlinear equation with a squashing property can be used. The transfer function is used to maintain the ANN's processed values within predefined limitations, ensuring that they are still adjustable. Only labelled data may be used to train ANNs. At the start of the training phase, the network's weights are set to very low, random values. The goal of training is to make systematic modifications to these weights such that the model generates consistent results across diverse types of data. The strength of ANNs lies in their ability to learn and map input-output correlations, allowing them to generalize this knowledge to fresh data. It is obvious that neural networks can learn a large range of input/output mappings in an efficient and resilient manner.

Because there are no hard and fast criteria for finding the appropriate number of neurons, connections, transfer functions, or squashing functions required to solve a particular problem, many consider ANN design to be a complex art form. Furthermore, ANNs frequently require a considerable amount of labelled training data, with a prominent heuristic proposing that the training dataset comprise at least five instances for each network link.

#### IV. EXPERIMENTAL RESULT ANALYSIS

The section analyses the outcome of the ML model with two different features.

##### A. Outcome of Histogram feature with ML Models

Three different algorithms—LR, NB, and ANN—were used to evaluate the integration of ML models with features of the FOH. The LR model correctly identified 268 leukemia and 289 healthy samples. There were 15 instances where healthy samples were incorrectly labelled as leukemia and 28 instances when leukemia samples were incorrectly labelled as healthy. The NB model was reliable, successfully classifying 285 leukemia and 292 healthy samples with just 11 false positives and 12 false negatives. The ANN model outperformed the others, predicting leukemia samples of 291 and healthy samples of 295 correctly and it wrongly predicted 9 healthy and 5 leukemia samples. Figure 2 is a confusion matrix depicting the model's performance on FOH features.

| MODEL | PREDICTED OUTCOME | ACTUAL OUTCOME |        |
|-------|-------------------|----------------|--------|
| LR    |                   | TP=268         | FP=15  |
|       |                   | FN=28          | TN=289 |
| NB    |                   | TP=285         | FP=11  |
|       |                   | FN=12          | TN=292 |
| ANN   |                   | TP=291         | FP=9   |
|       |                   | FN=5           | TN=295 |

Fig. 2. Confusion Matrix of FOH and ML models

Table 2 displays the model's performance metrics. ANN achieved the maximum accuracy of 97.67%, while LR achieved the lowest accuracy of 92.83%. Similarly, for additional criteria such as specificity, sensitivity, precision, and F1, ANN wins the race with values of 97.04%, 98.31%, 97%, and 97.65%, while NB comes in second and LR comes in last. Figure 3 depicts a comparison of ML model performance utilizing FOH features.

Table 2. Performance metrics of FOH and ML models

| ML Model    | LR    | NB    | ANN   |
|-------------|-------|-------|-------|
| Accuracy    | 92.83 | 96.17 | 97.67 |
| Specificity | 95.07 | 96.37 | 97.04 |
| Sensitivity | 90.54 | 95.96 | 98.31 |
| Precision   | 94.7  | 96.28 | 97    |
| F1          | 92.57 | 96.12 | 97.65 |

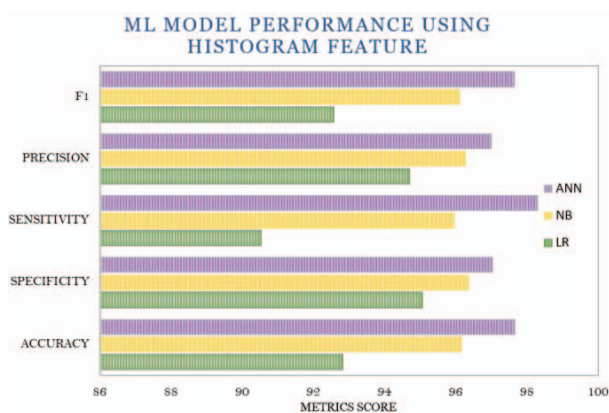


Fig. 3. Comparison chart of ML model by deploying FOH features

#### B. The outcome of GLCM feature with ML Models

The ML model's integration with GLCM features is assessed. When used together, the LR can correctly guess 273 samples with leukemia and 291 samples that are healthy. It identified 13 healthy samples and 23 leukemia samples incorrectly. The NB correctly predicted 293 leukemia and 295 healthy specimens. And it identified 7 healthy samples and 5 leukemia samples wrong. The ANN has successfully predicted 297 cases of leukemia and 298 cases of normal tissue. And it got 4 healthy samples and 1 leukemia sample incorrect. Figure 4 displays the obtained confusion matrix of the ML model on the GLCM features.

| MODEL | PREDICTED OUTCOME | ACTUAL OUTCOME |        |
|-------|-------------------|----------------|--------|
| LR    |                   | TP=273         | FP=13  |
|       |                   | FN=23          | TN=291 |
| NB    |                   | TP=293         | FP=7   |
|       |                   | FN=5           | TN=295 |
| ANN   |                   | TP=297         | FP=4   |
|       |                   | FN=1           | TN=298 |

Fig. 4. Confusion Matrix of GLCM and ML models

Table 3 shows the ML model's performance metrics by utilizing GLCM outcomes. ANN achieved an accuracy of 99.17%, whereas LR achieved no better than 94%. The results are the same across the board: ANN has the best specificity, sensitivity, precision, and F1, whereas NB comes in second and LR is bottom. Figure 5 displays a comparison of the ML models' results when GLCM features are used.

Table 3. Performance metrics of GLCM and ML models

| ML Model    | LR    | NB    | ANN   |
|-------------|-------|-------|-------|
| Accuracy    | 94    | 98    | 99.17 |
| Specificity | 95.72 | 97.68 | 98.68 |
| Sensitivity | 92.23 | 98.32 | 99.66 |
| Precision   | 95.45 | 97.67 | 98.67 |
| F1          | 93.81 | 97.99 | 99.17 |

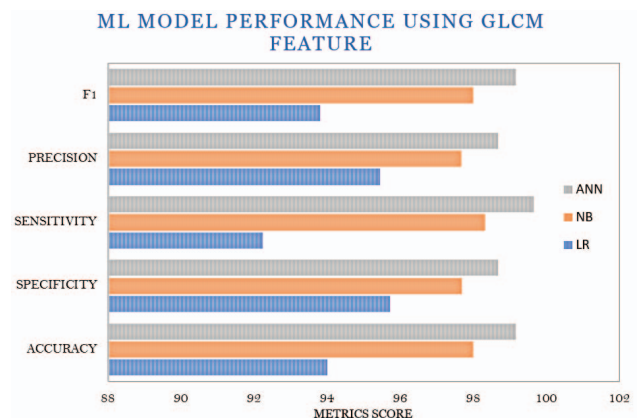


Fig. 5. Comparison chart of ML model by deploying GLCM features

#### V. CONCLUSION

This study described a diagnostic technique for ALL that uses an ML model to differentiate between leukemia and normal cells, hence assisting in real-world medical diagnosis. The C-NMC 2019 dataset served as the foundation for our research, and several pre-processing techniques were used to prepare the images for ML processing. Further, this study has tried six various combinations of ML models and feature extraction techniques. The following accuracies were obtained for each combination: 92.83% for FOH+LR, 96.17% for FOH+NB, 97.67% for FOH+ANN, 94% for GLCM+LR, 98% for

GLCM+NB, and 99.17% for GLCM+ANN. The simulation results showed that GLCM feature extraction paired with an ANN beat other models in terms of accuracy. This higher performance shows that it has the potential to considerably improve the diagnostic of ALL. The present focus of the study revolves around the identification of ALL. However, we are thinking about expanding our research in the future to cover additional forms of leukemia.

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